HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - II

Complex Variable - Integration

Topics covered in this unit :

- Line integral, contour integration
- Cauchy's integral theorem (simple case)
- Cauchy's integral formula
- Power series: Taylor's series, zeros of analytic functions
- Singularities, Laurent's series
- Residues, Cauchy residue theorem (without proof)
- Evaluation of definite integrals involving sine and cosine

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Line (Contour) Integral

1.1 Definition

Let C be a smooth curve from A to B given by $z = z(t) = x(t) + i\,y(t)$, $a <= t <= b$, and $f(z) = u + iv$ continuous on C. The line integral of f along C is:

$$\int_C f(z)\,dz = \int (u + iv)(dx + i\,dy) = \int (u\,dx - v\,dy) + i \int (v\,dx + u\,dy)$$

Parametric form

$$\int_C f(z)\,dz = \int_{t=a}^{b} f(z(t))\,z'(t)\,dt$$

- A contour is a piecewise smooth curve, a simple closed contour does not cross itself. Positive direction = anticlockwise.
- Integral over a closed contour is written as contour-integral (circle on the integral sign).

1.2 Properties

Property	Statement
Linearity	$\int (a f + b g)\,dz = a \int f\,dz + b \int g\,dz$
Reversal	$\int_{-C} f\,dz = -\int_C f\,dz$
Additivity	$\int_{C1+C2} = \int_{C1} + \int_{C2}$
ML inequality	$\left|\int_C f\,dz\right| <= M L$, where $\|f\| <= M$ on C and L = length of C

1.3 Useful parametrisations

Path	Parametrisation
Circle $\|z - a\| = r$	$z = a + r e^{i t}$, $0 <= t <= 2\pi$, $dz = i r e^{it}\,dt$
Line from z_1 to z_2	$z = z_1 + t(z_2 - z_1)$, $0 <= t <= 1$
Real axis segment	$z = x$, $dz = dx$
Parabola $y = x^2$	$z = x + i x^2$, $dz = (1 + 2 i x)\,dx$

Example 1

Evaluate $\int z^2\,dz$ from 0 to $1+i$ along the line $y = x$. Put $z = (1+i)t$, $dz = (1+i)\,dt$, $0 \le t \le 1$:

$\int = \int (1+i)^3 t^2\,dt = (1+i)^3/3 = (1+i)(2i)/3 = (2i-2)/3 = (-2+2i)/3$.

Example 2

Evaluate $\int \bar{z}\,dz$ from 0 to $1+i$ along $y = x$. $\bar{z} = (1-i)t$, $dz = (1+i)\,dt$: $\int = (1-i)(1+i)\int t\,dt = 2 \times 1/2 = 1$. Along a different path (x-axis then vertical) the value is $1+i$: depends on the path because $\bar{z}$ is not analytic.

Example 3 - the basic circle integral

> contour $\int$ over $|z-a| = r$ of $dz/(z-a)^n = 2\pi i$ $(n = 1)$; 0 $(n \neq 1$, integer$)$

Put $z - a = r\,e^{it}$: $dz = i\,r\,e^{it}\,dt$, so $\int = i\,r^{1-n}\int_0^{2\pi} e^{i(1-n)t}\,dt$. For $n = 1$ this is $i \times 2\pi = 2\pi i$; for $n \neq 1$ the integral of e^{ikt} over a full period is 0.

2. Cauchy's Integral Theorem

2.1 Statement (simple case)

Cauchy's Theorem (Cauchy-Goursat)

- If $f(z)$ is analytic at all points inside and on a simple closed contour C, then
- contour $\int_C f(z)\,dz = 0$
- Proof idea (Green's theorem): $\int (u\,dx - v\,dy) = \iint (-v_x - u_y)\,dA = 0$ and $\int (v\,dx + u\,dy) = \iint (u_x - v_y)\,dA = 0$ by C-R equations

2.2 Consequences

- Independence of path: if f is analytic in a simply connected region, Int from a to b of f dz depends only on a and b, not on the path
- Existence of antiderivative: Int from a to b f(z) dz = F(b) - F(a) where F' = f
- Deformation of contours: C1 can be deformed to C2 without changing the integral provided no singularity is crossed
- Multiply connected region (with holes): integral over outer C = sum of integrals over inner circles C1, C2, ...

Examples

(i) Int_C e^z dz = 0 for any closed C (e^z entire). (ii) Int_C z^3 dz = 0. (iii) Int_C dz/(z - 3) over |z| = 1 = 0 because singular point z = 3 lies outside C. (iv) Int_C 1/(z^2 + 4) dz over |z| = 1: singular points z = +/- 2i are outside |z| = 1 -> 0 .

3. Cauchy's Integral Formula

Cauchy's Integral Formula

- If f(z) is analytic inside and on a simple closed contour C and a is any point inside C, then
- f(a) = (1 / 2 pi i) contour Int_C f(z) / (z - a) dz
- i.e. contour Int_C f(z)/(z - a) dz = 2 pi i f(a)

Derivative form (Generalised formula)

f^(n)(a) = (n! / 2 pi i) contour Int_C f(z) / (z - a)^(n+1) dz

contour Int_C f(z)/(z - a)^(n+1) dz = 2 pi i f^(n)(a) / n!

The values of an analytic function inside a contour are determined by its values on the contour. Consequence: analytic function has derivatives of all

orders .

Example 1

Evaluate contour Int_C e^(2z)/(z - 1) dz, C: |z| = 2. Here a = 1 (inside C), f(z) = e^(2z). Integral = 2 pi i f(1) = 2 pi i e^2 .

Example 2 - using partial fractions

Evaluate contour Int_C dz/((z - 1)(z - 3)), C: |z| = 2. Partial fractions: 1/((z-1)(z-3)) = (1/2)[1/(z-3) - 1/(z-1)]. Only z = 1 is inside |z| = 2, so Int = (1/2)(0 - 2 pi i) = - pi i .

Example 3 - derivative form

Evaluate contour Int_C e^z/(z - 1)^3 dz, C: |z| = 3. Here n = 2, f(z) = e^z, f''(1) = e. Integral = 2 pi i f''(1)/2! = pi i e .

Example 4

Evaluate contour Int_C z/(z^2 + 1) dz, C: |z| = 2. Partial fractions: z/((z - i)(z + i)) = (1/2)[1/(z - i) + 1/(z + i)]. Both poles lie inside, so Int = (1/2)(2 pi i + 2 pi i) = 2 pi i .

4. Power Series - Taylor's Series

4.1 Sequences and series

A series Sum a_n (z - a)^n converges for |z - a| < R (radius of convergence R) and diverges for |z - a| > R. Ratio test: R = lim |a_n / a_(n+1)|.

4.2 Taylor's theorem

Taylor's Series

- If f(z) is analytic inside the circle C: |z - a| = R, then for all z inside C
- f(z) = f(a) + f'(a)(z - a) + f''(a)(z - a)^2 / 2! + ... = Sum (n = 0 to inf) f^(n)(a) (z - a)^n / n!
- When a = 0 it is called Maclaurin's series
- Radius of convergence = distance from a to the nearest singularity

4.3 Standard expansions (Maclaurin)

Function	Series	Valid for
e^z	1 + z + z^2/2! + z^3/3! + ...	all z
sin z	z - z^3/3! + z^5/5! - ...	all z
cos z	1 - z^2/2! + z^4/4! - ...	all z
sinh z	z + z^3/3! + z^5/5! + ...	all z
cosh z	1 + z^2/2! + z^4/4! + ...	all z
1/(1 - z)	1 + z + z^2 + z^3 + ...	|z| < 1
1/(1 + z)	1 - z + z^2 - z^3 + ...	|z| < 1
log(1 + z)	z - z^2/2 + z^3/3 - ...	|z| < 1
(1 + z)^n	1 + n z + n(n-1)/2! z^2 + ...	|z| < 1

Example 1 - expand f(z) = 1/(z + 2) about z = 1

Write z + 2 = 3 + (z - 1) = 3 [1 + (z - 1)/3]. Then f = (1/3) Sum (-1)^n ((z - 1)/3)^n valid for |z - 1| < 3 (nearest singularity z = -2 at distance 3).

Example 2 - expand sin z about z = pi/4

f = sin z, f' = cos z ...; f(pi/4) = f'(pi/4) = 1/sqrt2, f'' = - f ...; so sin z = (1/sqrt2)[1 + (z - pi/4) - (z - pi/4)^2/2! - (z - pi/4)^3/3! + ...].

4.4 Zeros of an analytic function

A point z = a is a zero of f(z) if f(a) = 0. It is a zero of order (multiplicity) m if f(a) = f'(a) = ... = f^(m-1)(a) = 0 but f^(m)(a) != 0, equivalently f(z) = (z - a)^m g(z), with g analytic and g(a) != 0.

Zero of order 1 is a simple zero . Zeros of a non-constant analytic function are isolated .

Function	Zero	Order
$z^2 - 1$	$z = +/- 1$	1 (simple)
$z^3 (z - 2)$	$z = 0$; $z = 2$	3 ; 1
sin z	$z = n\ pi$	1
$1 - \cos z$	$z = 0$	2 ($1 - \cos z = z^2/2 - \ldots$)
$z - \sin z$	$z = 0$	3 ($= z^3/6 - \ldots$)

5. Singularities

A point $z = a$ where $f(z)$ is not analytic is a singular point . If f is analytic in some neighbourhood of a except at a itself, a is an isolated singularity .

Type	Definition (from Laurent series)	Example
Removable	No negative powers; lim f(z) exists finite	sin z / z at z = 0 (value 1)
Pole of order m	Finite number (m) of negative powers; lim |f| = inf	$1/(z-1)^2$ at z = 1 (order 2)
Simple pole	Pole of order 1	$1/(z - 2)$ at z = 2
Essential	Infinite number of negative powers, limit does not exist	$e^{(1/z)}$, $\sin(1/z)$ at z = 0

How to find the order of a pole

- f(z) = phi(z)/(z - a)^m with phi(a) != 0 -> pole of order m
- Or: if f = g/h and h has a zero of order m at a while g(a) != 0 -> pole of order m
- Branch point (log z, sqrt z) and non-isolated singularities are other kinds
- Entire function: no finite singularity; singularity at infinity decided by f(1/w) at w = 0

Examples

(i) f = z/(z - 1)^2 (z + 2): poles z = 1 (order 2), z = -2 (simple). (ii) f = (1 - cos z)/z^2: z = 0 removable (limit 1/2). (iii) f = 1/sin z: simple poles at n pi. (iv) f = z e^(1/z): essential singularity at 0.

6. Laurent's Series

Laurent's theorem

- If f(z) is analytic in the annulus r1 < |z - a| < r2, then for all z in it
- f(z) = Sum (n = 0 to inf) a_n (z - a)^n + Sum (n = 1 to inf) b_n / (z - a)^n
- a_n = (1/2 pi i) contour Int f(z)/(z - a)^(n+1) dz ; b_n = (1/2 pi i) contour Int f(z)(z - a)^(n-1) dz
- First part = analytic (regular) part ; second part = principal part

- Principal part has no terms -> removable singularity; finite terms -> pole; infinite terms -> essential singularity.
- In practice we do not use the integrals: expand with standard series + binomial / geometric in the correct region.

Method

1. Mark singular points and draw the region (annulus) asked
2. Partial fractions if rational
3. For |z| < a use 1/(1 - z/a) = Sum (z/a)^n ; for |z| > a use 1/(z - a) = (1/z) Sum (a/z)^n
4. Collect powers of (z - a)

Example 1

Expand f(z) = 1/(z^2 - z) in (i) 0 < |z| < 1, (ii) |z| > 1. Partial fractions: 1/(z(z-1)) = 1/(z - 1) - 1/z.

(i) 0 < |z| < 1: 1/(z - 1) = - (1 + z + z^2 + ...), so f = - 1/z - 1 - z - z^2 - ...

(ii) |z| > 1: 1/(z - 1) = (1/z)(1 + 1/z + 1/z^2 + ...), so f = 1/z^2 + 1/z^3 + 1/z^4 + ...

Example 2 - f(z) = e^z / z^2 about z = 0

e^z = 1 + z + z^2/2! + ... so f = 1/z^2 + 1/z + 1/2 + z/6 + ... Pole of order 2; coefficient of 1/z = 1 (residue).

Example 3 - f(z) = 1/((z-1)(z-2)) in 1 < |z| < 2

f = 1/(z - 2) - 1/(z - 1). For |z| < 2: 1/(z - 2) = -(1/2) Sum (z/2)^n. For |z| > 1: 1/(z - 1) = Sum 1/z^(n+1).

f = - Sum (n=0 to inf) z^n / 2^(n+1) - Sum (n=0 to inf) 1/z^(n+1) .

Example 4 - sin z / z^3

= (z - z^3/6 + ...)/z^3 = 1/z^2 - 1/6 + z^2/120 - ... ; pole of order 2.

7. Residues and the Residue Theorem

7.1 Definition

If a is an isolated singularity of f with Laurent series about a, the residue of f at a is the coefficient b_1 of 1/(z - a) :

Res f(z) at z = a = b_1 = (1/2 pi i) contour Int_C f(z) dz

7.2 Formulae for residues

Case	Residue at z = a
Simple pole	lim (z -> a) (z - a) f(z)
f = P(z)/Q(z), Q(a) = 0, Q'(a) != 0	P(a) / Q'(a)
Pole of order m	(1/(m-1)!) lim (z -> a) d^(m-1)/dz^(m-1) [(z - a)^m f(z)]
Pole of order 2	lim d/dz [(z - a)^2 f(z)]
Removable singularity	0
Essential	Find coefficient of 1/(z - a) in Laurent series

Examples

(i) $f = 1/(z^2 + 1)$ at $z = i$: simple pole. Res = lim $(z - i)/((z - i)(z + i)) = 1/(2i)$.

(ii) $f = e^z/(z - 1)^2$ at $z = 1$: order 2. Res = $d/dz\ (e^z)$ at $1 = e$.

(iii) $f = z/(z^2 + 1)$... P/Q' rule: at $z = i$: $i/(2i) = 1/2$. (iv) $f = \cot z$ at $z = 0$: $\cos z / (\sin z)' = \cos 0/\cos 0 = 1$.

(v) $f = 1/(z^3 (z - 1))$ at $z = 0$ order 3: $(1/2!)\ d^2/dz^2\ [1/(z - 1)]$ at $0 = (1/2)(2/(z-1)^3) = -1$.

7.3 Cauchy's Residue Theorem (without proof)

Residue Theorem

- If f(z) is analytic inside and on a simple closed contour C except for a finite number of isolated singularities a1, a2, ..., an inside C, then
- contour Int_C f(z) dz = 2 pi i x (sum of residues of f at a1, ..., an)

Example 1

Evaluate contour Int_C $(z + 1)/(z^2 (z - 2))\ dz$, C: $|z| = 3$ (both poles inside). Res at $z = 2$ (simple): $(2 + 1)/4 = 3/4$. Res at $z = 0$ (order 2): $d/dz\ [(z+1)/(z - 2)]$ at $0 = [(z - 2) - (z + 1)]/(z - 2)^2 = -3/4$. Sum = 0, so integral = 0.

Example 2

contour Int_C dz/(z^2 + 4) over |z - i| = 2: pole z = 2i is inside (distance |2i - i| = 1 < 2), z = -2i is outside. Res at 2i = 1/(4i). Integral = 2 pi i (1/4i) = pi/2 .

Example 3

contour Int_C tan z dz over |z| = 2: poles of tan z at z = +/- pi/2 (|pi/2| = 1.57 < 2). Res of sin z/cos z at pi/2 = sin(pi/2)/(-sin(pi/2)) = -1. Same for -pi/2. Integral = 2 pi i (-2) = -4 pi i .

8. Definite Integrals Involving sin and cos

Type: Int (0 to 2 pi) F(cos theta, sin theta) d theta , F a rational function. Substitute z = e^(i theta) - theta from 0 to 2 pi means z goes once round the unit circle |z| = 1.

Substitution

- z = e^(i theta), dz = i z d theta, d theta = dz/(i z)
- cos theta = (z + 1/z)/2 , sin theta = (z - 1/z)/(2i)
- cos n theta = (z^n + z^-n)/2 ; sin n theta = (z^n - z^-n)/2i
- Integral = contour Int over |z| = 1 of G(z) dz = 2 pi i x Sum of residues inside |z| = 1

Example 1

Evaluate I = Int (0 to 2pi) d theta/(5 + 4 cos theta).

5 + 4 cos t = 5 + 2(z + 1/z) = (2z^2 + 5z + 2)/z = 2 (z + 2)(z + 1/2)/z. I = contour Int dz/(i z) x z/[2 (z + 2)(z + 1/2)] = (1/2i) contour Int dz/((z + 2)(z + 1/2)).

Poles: z = -2 (outside), z = -1/2 (inside). Res at -1/2 = 1/(-1/2 + 2) = 2/3. I = (1/2i)(2 pi i)(2/3) = 2 pi/3 .

Example 2

Evaluate Int (0 to 2pi) d theta/(a + b cos theta), a > b > 0. Standard result:

check: a = 5, b = 4 gives 2 pi/3 OK

Int (0 to 2 pi) d theta / (a + b cos theta) = 2 pi / sqrt(a^2 - b^2)

Example 3

Int (0 to 2pi) d theta/(2 + cos theta) = 2 pi / sqrt(4 - 1) = 2 pi/sqrt3 .

Example 4 - with cos 2 theta

I = Int (0 to 2pi) cos 2t /(5 - 4 cos t) dt. Take I = Re Int e^(2it)/(5 - 4 cos t) dt. With z = e^(it): 5 - 4cos t = (-2z^2 + 5z - 2)/z = -2(z - 2)(z - 1/2)/z. So Int = contour Int z^2 . z . dz /(i z . (-2)(z - 2)(z - 1/2)) = (-1/2i) contour Int z^2 dz/((z - 2)(z - 1/2)).

Only z = 1/2 inside: Res = (1/4)/(1/2 - 2) = -1/6. Integral = (-1/2i)(2 pi i)(-1/6) = pi/6 .

Quick revision - Unit II

- Int_C f dz = Int f(z(t)) z'(t) dt, circle: contour Int dz/(z - a) = 2 pi i
- Cauchy: analytic => integral = 0, formula: f(a) = (1/2 pi i) Int f/(z - a) dz
- Taylor: Sum f^(n)(a)(z - a)^n / n!, Laurent: analytic part + principal part
- Residue (simple pole) = lim (z - a) f(z); Residue theorem: Int = 2 pi i x Sum Res
- Trig integrals: z = e^(i theta), d theta = dz/(iz), poles inside |z| = 1